

Maintenance Schedule

Gas Engine

20V4000S24N

Application group 4A

MS50288/00E



Power. Passion. Partnership.

Engine model	kW/cyl.	Application group
20V4000S24N	130 kW/cyl.	4A, continuous operation, unrestricted

Table 1: Applicability

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1 Maintenance Schedule

1.1 Preface

In the case of emission-certified products, nonobservance of the maintenance instructions can result in violation of the statutory specifications. Modification or removal of mechanical or electronic components or the installation of additional mechanical or electronic components as well as the execution of calibration processes that might affect the emission characteristics of the engine are prohibited by emission regulations. Components relevant to emissions may only be serviced, replaced or repaired if components approved by MTU are used. Components relevant to emissions include e.g. control units, data records, sensors, injectors, exhaust flaps and all components of exhaust after-treatment systems.

The maintenance system for MTU products is based on a preventive maintenance concept. Preventive maintenance facilitates advance operational planning and ensures a high level of equipment availability. The maintenance intervals as well as the required scopes of maintenance tasks are based on operational experience and therefore to be considered as recommendations. Additional maintenance work and/or changes to the maintenance intervals may be required in the case of difficult operational and ambient conditions.

Adherence to the specified maintenance intervals is essential to maintain product safety.

The intervals according to which the maintenance tasks have to be carried out are specified as operating hours and time limits. Whichever value is reached first shall apply.

The individual maintenance intervals are assigned to the qualification levels QL1 to QL4.

QL1: Operational monitoring and maintenance work which does not require disassembly of the product.

QL2: Exchange of components and parts (corrective only, not part of the maintenance schedule).

QL3: Maintenance work which requires partial disassembly of the product.

QL4: Maintenance work which requires complete disassembly of the product.

Additional notes on maintenance and preservation:

For the change intervals for fluids and lubricants and for information on preservation, refer to the relevant specifications of the component manufacturers and the MTU Fluids and Lubricants Specifications. The latest version for drive systems is available at: <http://www.mtu-online.com>, for power generation at: <http://www.mtuonsiteenergy.com>.

Components which are not listed in this maintenance schedule must be serviced in accordance with Manufacturer's Instructions.

This maintenance schedule does not apply to engines with an EPA/CARB emissions certificate.

1.2 Additional notes on the maintenance of gas engines

General information

The maintenance schedule matrix normally finishes with extended component maintenance or with engine overhaul. Following this, maintenance work is to be continued at the intervals indicated.

The numbers stated in the task list serve as a reference for the scope of parts required.



Repairs tasks on the gas engine, for which the gas-conducting area is opened, and leak or stress tests must only be carried out by specialized companies. As a rule these are companies that are approved for the gas industry with qualified personnel who have been trained for technical gas applications.



Specifications for fluids and lubricants, guideline values for their maintenance and change intervals and lists of recommended fluids and lubricants are contained in the MTU Fluids and Lubricants Specifications and in the fluids and lubricants specifications produced by the component manufacturers. They are therefore not included in the maintenance schedule (exception: deviations from the Fluids and Lubricants Specifications). All fluids and lubricants used must meet MTU specifications and be approved by the relevant component manufacturer.

Among other items, the operator/customer must carry out the following additional maintenance work:

- Protect components made of rubber or synthetic material from oil. Never treat them with organic detergents. Wipe with a dry cloth only.
- Gas filter:
The maintenance interval depends on gaseous fuel quality (purity). Fuel filters upstream of the engine must be regularly cleaned and renewed if necessary.
- Battery:
Battery maintenance depends on duty and ambient conditions. The specifications of the battery manufacturer apply.

Items listed in this maintenance schedule but which are not included in the MTU scope of supply may be disregarded.

Maintenance of non-MTU components

Please refer to the Operating Instructions provided by the relevant component manufacturer for the intervals for these maintenance tasks.

Maintenance tasks

The respective maintenance tasks to be carried out are based on operating hours (see the following pages).

1.3 Checks by the operator



Always observe the safety instructions.



Non-observance of these instructions may lead to damage to the system!



When the object of delivery is used abroad or in other special regions, MTU Friedrichshafen GmbH is not responsible for observance of the statutory and other specifications / standards applicable at the place of use.

Recommended daily checks – System

- Check exhaust gas piping for condensate leaks (visual inspection)
- Check condensate collecting tank downstream of condensate drain (if fitted)
- Carry out visual inspection of engine and plant for leaks / damage (bellows, hoses etc.)
- Check gas supply / gas system for damage (visual inspection)
- Check fleece of air filter for contamination, replace if necessary
- Oil temperature
- Oil pressure
- Leaks
- Oil level
- Noise, vibrations

Recommended daily checks – Control system

The following measured values and functions are monitored by the control system to protect the plant. These messages can also be called up via remote access (option) and can be monitored by the customer in the display.

- Power
- Lube oil level
- Lube oil pressure
- Coolant temperature
- Coolant pressure
- Battery voltage
- Smooth running of engine
- Exhaust gas temperature
- Engine coolant supply and return temperature (option)
- Oil bleed system function (option)

Recommended weekly checks

- Check air filter box for cracks
- Check air filter box for leak-tight seating on engine
- Check air filter for leak-tight seating on air filter box
- Lube oil consumption or content of fresh oil tank
- Engine room temperature
- Check of operating hours
- Check of number of starts
- General visual inspection of module / engine-generator set

Additional checks

The operator's gas installation (upstream of twin solenoid valve) must be inspected for leaks at the following intervals:

- Every year for engines with turbocharger

The following checks are to be carried out at appropriate intervals, depending on the degree of contamination:

- Check room air fan and fresh air filter for contamination
- Check filter or air-conditioning unit for control cabinet for contamination
- Check recoler for contamination

Electrical systems

Electrical systems and operating equipment must be checked prior to initial commissioning, after repairs, prior to re-commissioning and at regular intervals. The intervals are to be so arranged that defects, which are to be expected, are detected at an early stage.

Here, among others, the following requirements apply:

- Accident prevention regulations BGV A3 "Electrical Systems and Operating Equipment"
- Technical rules for operational safety TRBS 2131 "Electrical Hazards"
- Industrial Safety Directive (BetrSichV)
- DIN VDE 0701/0702 Repairs, modifications and testing of electrical equipment/repeat tests on electrical units
- DIN EN 60204-1 Safety of machinery - Electrical equipment of machines

The operating company has the responsibility to see that these tests are carried out correctly. Electrical systems and fixed electrical operating equipment must be tested at appropriate intervals by a qualified electrician for their correct condition. The test is to include the following:

- Visual inspection and testing of protective equipment
- Testing of protective conductor resistance, insulation resistance, leakage current replacement, protective conductor current and differential current
- Carrying out fault diagnoses
- Preparation of test reports and inspection stickers

The industrial safety regulations, the technical rules for operational safety and the BGV A3 are fundamental with regard to adherence to the protective measures for "Electrical Systems and Operating Equipment". The operating company is accordingly obligated to care for the safety of electrical systems and operating equipment.

	Designation
DIN	Deutsches Institut für Normung (Federal German Standards Institute)
EN	Europäische Normung (European Standards)
ISO	International Standards Organization
VDE	Verband der Elektrotechnik, Elektronik und Informationstechnik (Association for electrical engineering, electronics and information technology)
BGV	Berufsgenossenschaftliche Vorschriften (Trade association regulations)
TRBS	Technische Regeln für Betriebssicherheit (Technical rules for operational safety)

Table 2: Designations for standards and regulations



These regulations are contained in the manufacturers' instructions, legal requirements and technical guidelines valid in the individual countries. Since there can be major differences from country to country, it is not possible to provide a universally applicable statement on the rules to be observed within the framework of this maintenance schedule.

The user of the product mentioned herein is obligated to inform him-/herself about the provisions applicable in his/her country. MTU Friedrichshafen GmbH accepts no liability whatsoever for improper or illegal use of the electrical system, which it has approved.

1.4 Noncyclical maintenance tasks

No.	Qualification	Task	Maintenance tasks	Reference task	Offset	Value	Unit	Option
0010	QL1	WM00781	Clean gas filter. Initial cleaning after 100 operating hours		0	100	HRS	
0020	QL1	WM00094	Check valve clearance, adjust if necessary. ATTENTION! First adjustment after 1,000 operating hours on a new engine and after 1,000 operating hours following each cylinder head overhaul		0	1000	HRS	
Reference task: Name of the maintenance task the offset refers to Offset: Interval of a maintenance task which depends on another interval Value: Fixed value for binding execution of a maintenance task								

1.5 Maintenance Schedule Matrix 63000 HRS

0 to 27000 Operating hours

Task	Limit	Operating hours [HRS]																													
		Daily	100	1000	2000	3000	4000	5000	6000	7000	8000	9000	10000	10500	11000	12000	13000	14000	15000	16000	17000	18000	19000	20000	21000	22000	23000	24000	25000	26000	27000
WM00428	1 WK																														
WM00296	1 MON																														
WM00286	-	X																													
WM00427	-	X																													
WM00426	-	X																													
WM00779	18 YR				X		X		X		X			X		X		X		X		X		X		X		X		X	
WM00511	1 YR					X		X		X				X		X		X		X		X		X		X		X		X	
WM00050	1 YR					X		X		X				X		X		X		X		X		X		X		X		X	
WM00136	1 YR					X		X		X				X		X		X		X		X		X		X		X		X	
WM00509	2 YR					X		X		X				X		X		X		X		X		X		X		X		X	
WM00067	2 YR					X		X		X				X		X		X		X		X		X		X		X		X	
WM00510	3 YR					X		X		X				X		X		X		X		X		X		X		X		X	
WM00076	3 YR					X		X		X				X		X		X		X		X		X		X		X		X	
WM00126	6 MON					X		X		X				X		X		X		X		X		X		X		X		X	
WM00780	18 YR					X		X		X				X		X		X		X		X		X		X		X		X	
WM00434	18 YR					X		X		X				X		X		X		X		X		X		X		X		X	
WM00094	18 YR			X		X		X		X				X		X		X		X		X		X		X		X		X	
WM00150	18 YR					X		X		X				X		X		X		X		X		X		X		X		X	
WM00433	3 YR							X						X				X				X				X					
WM00098	18 YR									X										X											X
WM00573	6 YR											X													X						
WM00447	6 YR																								X						
WM00431	18 YR																								X						
WM00792	6 YR																														
WM00442	18 YR											X													X						
WM00783	4 YR																								X						
WM00108	18 YR																								X						
WM00430	18 YR																								X						
WM00444	18 YR																								X						
WM00099	18 YR																								X						
WM00093	18 YR																								X						
WM00082	18 YR																								X						
WM00443	18 YR																								X						
WM00782	18 YR																								X						
WM00448	18 YR																								X						
WM00005	18 YR																								X						
WM00439	18 YR																														
WM00097	18 YR																														
WM00001	18 YR																														
WM00002	18 YR																														
WM00111	18 YR																														

WK = Weeks

MON = Months

YR = Years

HRS = Hours

Task	Limit	Operating hours [HRS]																													
		Daily	100	1000	2000	3000	4000	5000	6000	7000	8000	9000	10000	10500	11000	12000	13000	14000	15000	16000	17000	18000	19000	20000	21000	22000	23000	24000	25000	26000	27000
WM00104	18 YR																														
WM00103	18 YR																														
WM00095	18 YR																														
WM00092	18 YR																														
WM00070	18 YR																														
WM00127	18 YR																														
WM00137	18 YR																														
WM00781	1 YR		X	X	X	X	X	X	X	X	X	X		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
WM00464	2 YR					X			X			X			X			X			X			X			X				X
WM00499	1 YR								X						X						X							X			
WM00383	3 YR											X									X										X
WK = Weeks																															
MON = Months																															
YR = Years																															
HRS = Hours																															

28000 to 55000 Operating hours

Task	Limit	Operating hours [HRS]																													
		28000	29000	30000	31000	31500	32000	33000	34000	35000	36000	37000	38000	39000	40000	41000	42000	43000	44000	45000	46000	47000	48000	49000	50000	51000	52000	52500	53000	54000	55000
WM00428	1 WK																														
WM00296	1 MON																														
WM00286	-																														
WM00427	-																														
WM00426	-																														
WM00779	18 YR	X		X		X		X		X		X		X		X		X		X		X		X		X		X		X	
WM00511	1 YR			X			X			X			X			X			X			X			X				X		
WM00050	1 YR			X			X			X			X			X			X			X			X				X		
WM00136	1 YR			X			X			X			X			X			X			X			X				X		
WM00509	2 YR			X			X			X			X			X			X			X			X				X		
WM00067	2 YR			X			X			X			X			X			X			X			X				X		
WM00510	3 YR			X			X			X			X			X			X			X			X				X		
WM00076	3 YR			X			X			X			X			X			X			X			X				X		
WM00126	6 MON			X			X			X			X			X			X			X			X				X		
WM00780	18 YR			X			X			X			X			X			X			X			X				X		
WM00434	18 YR			X			X			X			X			X			X			X			X				X		
WM00094	18 YR			X			X			X			X			X			X			X			X				X		
WM00150	18 YR			X			X			X			X			X			X			X			X				X		
WM00433	3 YR			X						X				X				X				X							X		
WM00098	18 YR									X								X												X	
WM00573	6 YR				X									X													X				
WM00447	6 YR													X																	
WM00431	18 YR													X																	
WM00792	6 YR					X																									
WM00442	18 YR				X									X													X				
WK = Weeks																															
MON = Months																															
YR = Years																															
HRS = Hours																															

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Task	Limit	Operating hours [HRS]																													
		28000	29000	30000	31000	31500	32000	33000	34000	35000	36000	37000	38000	39000	40000	41000	42000	43000	44000	45000	46000	47000	48000	49000	50000	51000	52000	52500	53000	54000	55000
WM00783	4 YR																X														
WM00108	18 YR																X														
WM00430	18 YR																X														
WM00444	18 YR																X														
WM00099	18 YR																X														
WM00093	18 YR																X														
WM00082	18 YR																X														
WM00443	18 YR																X														
WM00782	18 YR																X														
WM00448	18 YR																X														
WM00005	18 YR																X														
WM00439	18 YR					X																									
WM00097	18 YR																														
WM00001	18 YR																														
WM00002	18 YR																														
WM00111	18 YR																														
WM00104	18 YR																														
WM00103	18 YR																														
WM00095	18 YR																														
WM00092	18 YR																														
WM00070	18 YR																														
WM00127	18 YR																														
WM00137	18 YR																														
WM00781	1 YR	X	X	X	X		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X		X	X	X
WM00464	2 YR			X				X			X			X			X			X			X			X				X	
WM00499	1 YR			X							X						X					X								X	
WM00383	3 YR										X									X										X	
WK = Weeks																															
MON = Months																															
YR = Years																															
HRS = Hours																															

56000 to 63000 Operating hours

Task	Limit	Operating hours [HRS]																													
		56000	57000	58000	59000	60000	61000	62000	63000																						
WM00428	1 WK																														
WM00296	1 MON																														
WM00286	-																														
WM00427	-																														
WM00426	-																														
WM00779	18 YR	X		X		X		X																							
WM00511	1 YR		X			X			X																						
WM00050	1 YR		X			X			X																						
WM00136	1 YR		X			X			X																						
WK = Weeks																															
MON = Months																															
YR = Years																															
HRS = Hours																															

Task	Limit	Operating hours [HRS]																											
		56000	57000	58000	59000	60000	61000	62000	63000																				
WM00509	2 YR		X			X			X																				
WM00067	2 YR		X			X			X																				
WM00510	3 YR		X			X			X																				
WM00076	3 YR		X			X			X																				
WM00126	6 MON		X			X			X																				
WM00780	18 YR		X			X			X																				
WM00434	18 YR		X			X			X																				
WM00094	18 YR		X			X			X																				
WM00150	18 YR		X			X			X																				
WM00433	3 YR					X																							
WM00098	18 YR								X																				
WM00573	6 YR								X																				
WM00447	6 YR								X																				
WM00431	18 YR								X																				
WM00792	6 YR								X																				
WM00442	18 YR								X																				
WM00783	4 YR								X																				
WM00108	18 YR								X																				
WM00430	18 YR								X																				
WM00444	18 YR								X																				
WM00099	18 YR								X																				
WM00093	18 YR								X																				
WM00082	18 YR								X																				
WM00443	18 YR								X																				
WM00782	18 YR								X																				
WM00448	18 YR								X																				
WM00005	18 YR								X																				
WM00439	18 YR								X																				
WM00097	18 YR								X																				
WM00001	18 YR								X																				
WM00002	18 YR								X																				
WM00111	18 YR								X																				
WM00104	18 YR								X																				
WM00103	18 YR								X																				
WM00095	18 YR								X																				
WM00092	18 YR								X																				
WM00070	18 YR								X																				
WM00127	18 YR								X																				
WM00137	18 YR								X																				
WM00781	1 YR	X	X	X	X	X	X	X	X																				
WM00464	2 YR		X			X			X																				
WM00499	1 YR					X																							
WM00383	3 YR								X																				
WK = Weeks MON = Months YR = Years HRS = Hours																													

1.6 Cyclical maintenance tasks 63000 HRS

Qualification	Interval [HRS]	Limit	Item	Maintenance tasks	Option	Task
Engine						
QL1	-	1 WK	ENGINE IN GENERAL	Check engine oil level in storage tank		WM00428
QL1	-	1 MON	ENGINE IN GENERAL	Test run at not below 1/3 load and at least until steady-state temperature is reached (monthly)		WM00296
QL1	24	-	ENGINE IN GENERAL	Carry out visual inspection of engine for general condition and leaks		WM00286
QL1	24	-	ENGINE IN GENERAL	Check for abnormal running noise and vibrations		WM00427
QL1	24	-	ENGINE IN GENERAL	Check operating room for gas smell		WM00426
QL1	2000	18 YR	IGNITION SYSTEM (FOR GAS ENGINES)	Replace spark plugs TA-Luft/2 and seal in spark-plug connector	X	WM00779
QL1	3000	1 YR	ENGINE IN GENERAL	Check emissions values, adjust if required. Observe local emissions regulations		WM00511
QL1	3000	1 YR	ENGINE MOUNTING	Check general condition of resilient mounts (visual inspection)		WM00050
QL1	3000	1 YR	ENGINE MOUNTING	Check screws for strong fit		WM00136
QL1	3000	2 YR	IGNITION SYSTEM (FOR GAS ENGINES)	When exchanging spark plug: Clean and check spark plug connector, replace if required		WM00509
QL1	3000	2 YR	OIL FILTRATION/COOLING	Fit new engine oil filters each time the engine oil is changed or, at the latest, on expiry of the time limit (given in years)		WM00067
QL1	3000	3 YR	CRANKCASE BREATHER	Check crankcase pressure control, adjust if required		WM00510
QL1	3000	3 YR	AIR FILTER	Replace air filter		WM00076
QL1	3000	6 MON	ENGINE MOUNTING	Measure height of rubber element		WM00126
QL1	3000	18 YR	IGNITION SYSTEM (FOR GAS ENGINES)	Replace spark plugs TA-Luft and seal in spark-plug connector	X	WM00780
QL1	3000	18 YR	ENGINE IN GENERAL	Check ignition timing, adjust if required		WM00434
QL1	3000	18 YR	VALVE DRIVE	Check valve clearance, adjust if necessary. ATTENTION! First adjustment after 1,000 operating hours on a new engine and after 1,000 operating hours following each cylinder head overhaul		WM00094
QL1	3000	18 YR	EXHAUST TEMPERATURE MONITORING SYSTEM	Fit new NOx sensor		WM00150
QL1	6000	3 YR	GAS PIPEWORK (FOR GAS ENGINES)	Check lines carrying gas for leaks, seal if required		WM00433
QL1	9000	18 YR	CYLINDER HEAD	Measure valve protrusion		WM00098
QL1	10500	6 YR	CRANKCASE BREATHER	Replace fine filter		WM00573
QL1	21000	6 YR	CRANKCASE BREATHER	Change coarse filter (coalescence filter)		WM00447
QL1	21000	18 YR	ELECTRIC STARTING	Replace starter		WM00431
QL1	31500	6 YR	COOLANT SUPPLY SYSTEM	Replace flexible hose lines for engine coolant connection		WM00792
QL3	10500	18 YR	EXHAUST TURBOCHARGER, GENERAL OR H.P. TURBOCHARGER	Clean exhaust turbocharger		WM00442
QL3	21000	4 YR	ACCESSORIES FOR ENGINE GOVERNOR (ELECTRONIC) CONTROL SYSTEM	Replace humidity sensor		WM00783
QL3	21000	18 YR	CRANKCASE	Replace cylinder liners		WM00108
QL3	21000	18 YR	PISTON	Replace piston		WM00430
QL3	21000	18 YR	CONNECTING ROD	Replace conrod		WM00444
QL3	21000	18 YR	CONNECTING ROD	Replace conrod bearings		WM00099
QL3	21000	18 YR	VALVE DRIVE	Check rocker arms and valve bridge for wear Inspect swing follower through pushrod bore using endoscope		WM00093
QL3	21000	18 YR	EXHAUST TURBOCHARGER, GENERAL OR H.P. TURBOCHARGER	Overhaul exhaust turbocharger		WM00082

WK = Weeks
 MON = Months
 HRS = Hours
 YR = Years

Qualification	Interval [HRS]	Limit	Item	Maintenance tasks	Option	Task
QL3	21000	18 YR	GAS-MIXTURE COOLING (FOR GASENGINES)	Check for contamination and leaks, clean if required		WM00443
QL3	21000	18 YR	AIR SUPPLY TO CYLINDERS	Check for contamination, clean if necessary	X	WM00782
QL3	21000	18 YR	COOLANT CIRCUIT, GENERAL, HT CIRCUIT	Replace O-rings in connectors between coolant distribution housing and mixture cooler		WM00448
QL3	21000	18 YR	ENGINE IN GENERAL	Replace rubber bellows		WM00005
QL3	31500	18 YR	CRANKSHAFT, PTO ON AUX PTO END	Replace vibration damper		WM00439
QL3	63000	18 YR	CYLINDER HEAD	Overhaul cylinder head		WM00097
QL4	63000	18 YR	ENGINE IN GENERAL	Replace elastomeric parts and seals		WM00001
QL4	63000	18 YR	ENGINE IN GENERAL	Disassemble engine completely. Inspect engine components acc. to assembly instructions; repair or replace as necessary		WM00002
QL4	63000	18 YR	CRANKCASE	Replace camshaft bearings		WM00111
QL4	63000	18 YR	GEARCASE, AUX PTO END	Gear train: Check tooth flanks for wear (visual inspection), replace bushing		WM00104
QL4	63000	18 YR	CRANKSHAFT WITH BEARINGS	Replace crankshaft bearings		WM00103
QL4	63000	18 YR	CAMSHAFT	Replace thrust bearing flange		WM00095
QL4	63000	18 YR	VALVE DRIVE	Replace swing follower and swing follower shaft		WM00092
QL4	63000	18 YR	LUBE OIL PUMP AND DRIVE	Check pump, replace if necessary		WM00070
QL4	63000	18 YR	ENGINE MOUNTING	Replace rubber elements of engine mounts		WM00127
QL4	63000	18 YR	WIRING, GENERAL, FOR ENGINE/GEARBOX/PLANT	Replace wiring harness		WM00137
Accessories						
QL1	1000	1 YR	COMPONENTS GAS / FUEL SYSTEM PLANT / CONTAINER	Clean gas filter. Initial cleaning after 100 operating hours		WM00781
QL1	3000	2 YR	COMPONENTS GAS / FUEL SYSTEM PLANT / CONTAINER	Check gas filter, replace if necessary		WM00464
QL1	6000	1 YR	COMPONENTS GAS / FUEL SYSTEM PLANT / CONTAINER	Check function of valve sealing and check shut-off device		WM00499
QL1	9000	3 YR	COMPONENTS GAS / FUEL SYSTEM PLANT / CONTAINER	Fit new main gas filter		WM00383
WK = Weeks MON = Months HRS = Hours YR = Years						